№ 3.47

Дано:

$t_к = 21\,°C$

$t_{y_1} = 26\,°C$

$t_{y_2} = 31\,°C$

Найти: $\frac{A_2}{A_1}$

Решение:

$$A = \eta Q_н = \eta \alpha \Delta t = \left(1 - \frac{T_х}{T_н}\right)\alpha \Delta T$$

$$\frac{A_1}{A_2} = \frac{\left(1 - \frac{t_к}{t_{y_1}}\right)\alpha \Delta t_1}{\left(1 - \frac{t_к}{t_{y_2}}\right)\alpha \Delta t_2} \Rightarrow \frac{A_2}{A_1} \approx 4$$

№ 4.80

Дано:

$n = 2$

$H = 12{,}2$ км

$g = 8{,}87$ м/с^2

$C_v = 5R$

Найти: $T_п$

Решение:

$$dp = -\rho g\, dh, \quad pV^{\gamma} = const$$

$$pV = \frac{m}{M}RT \Rightarrow \rho = \frac{pM}{RT}; \quad V = \frac{\nu RT}{p}$$

$$p = \rho \frac{RT}{M}$$

1) $dp = -\frac{pM}{RT}\, dhg$; 2) $p \cdot \left(\frac{\nu RT}{p}\right)^{\gamma} = \alpha$

$$\frac{dp}{p} = -\frac{M}{RT}\, dh\, g$$

$$p^{\gamma-1} \cdot (\nu R)^{\gamma} \cdot T^{\gamma} = \alpha$$

$$(\gamma-1)\ln p + C_1 + \gamma \ln T = C_2$$

$$(\gamma-1)\frac{dp}{p} + \gamma \frac{dT}{T} = 0$$

$$\frac{dp}{p} = -\frac{\gamma}{\gamma-1}\frac{dT}{T}$$

$$\frac{\gamma}{\gamma-1}\frac{dT}{\cancel{T}} = \frac{M}{R\cancel{T}}\, dhg$$

$$\frac{\gamma}{\gamma-1}\, dT = \frac{M}{R}\, dhg$$

$$C_p\,(T - T_н) = MgH$$

$$T = T_н + \frac{MgH}{C_p}$$

со знаками ?

3) $\ln p = -\ln\rho - \ln T + C$

$$\begin{cases} \dfrac{dp}{p} = -\dfrac{d\rho}{\rho} - \dfrac{dT}{T} \\ \dfrac{dp}{p} = \dfrac{-\gamma}{\gamma-1}\dfrac{dT}{T} \end{cases} \qquad -\frac{d\rho}{\rho} - \frac{dT}{T} = \frac{-\gamma}{\gamma-1}\frac{dT}{T}$$

$$\frac{c_V}{R} = \frac{c_V}{c_p - c_V} = \frac{1}{\gamma-1}$$

$$\frac{d\rho}{\rho} = -\frac{dT}{T}\,\frac{1}{\gamma-1}$$

$$\ln\frac{\rho_к}{\rho_н} = \ln\frac{T_н}{T_к}\,\frac{1}{\gamma-1}$$

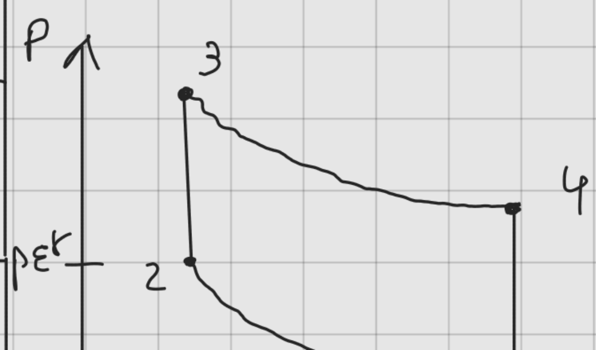
$$h = \left(\frac{\rho_к}{\rho_н}\right)^{\frac{R}{c_V}} = \frac{T_н}{T_к} =$$

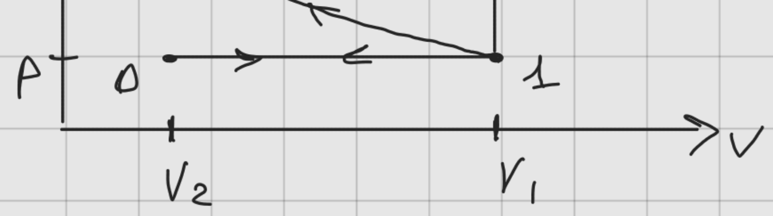
$$\left(\frac{\rho_к}{\rho_н}\right)^{\frac{R}{c_V}} = 1 : \frac{T + \frac{Mgh}{c_p}}{T}$$

$$T + \frac{Mgh}{c_p} = T\left(\frac{\rho_к}{\rho_н}\right)^{\frac{R}{c_V}}$$

$$T = \frac{Mgh}{c_p\left(-\left(\frac{\rho_к}{\rho_н}\right)^{\frac{R}{c_V}} + 1\right)} \approx 740\,К$$

№3.50

Дано:	Решение:
P_0, V_2, V_1 $\varepsilon = \frac{V_1}{V_2} = 5$ $\gamma = \frac{4}{3}$	
Найти: η	



$$1)\ p \cancel{V_2^\gamma} \cdot \varepsilon^\gamma = p_2 \cancel{V_2^\gamma} \Rightarrow p_2 = p\varepsilon^\gamma$$

$$p_3 = \alpha p \varepsilon^\gamma$$

$$\alpha p \cancel{\varepsilon^\gamma} \cancel{V_2^\gamma} = \cancel{V_2^\gamma} \cancel{\varepsilon^\gamma} p_4 \Rightarrow p_4 = \alpha p$$

$$\eta = 1 - \frac{\cancel{\tfrac{3}{2}} V_1 (\alpha - 1) p}{\cancel{\tfrac{3}{2}} V_2 \cdot (\alpha p \varepsilon^\gamma - p \varepsilon^\gamma)} = 1 - \frac{V_1}{V_2} \frac{1}{\varepsilon^\gamma} \approx 0{,}42$$

№4.58

Дано:

$pV^n = const$

$V_1 \to V_2$

$n = 3,\ V_1 = 1,\ V_2 = 3$

$p_1 = 20$ атм

$C_v = \frac{3R}{2}$

Найти: ΔU, ΔS, Q

Решение:

$$1)\ \Delta U = C_v (T_2 - T_1)$$

$$\nu R T V^{n-1} = const$$

$$T = \frac{const}{V^{n-1} \nu R}$$

$$\Rightarrow \boxed{\Delta U = C_v \frac{pV^n}{R} \left(\frac{1}{V_2^{n-1}} - \frac{1}{V_1^{n-1}} \right)}$$

$$S = S_0 + C_v \ln \frac{T}{T_0} + R \ln \frac{V}{V_0}$$

$$pV = RT$$

$$T = \frac{pV}{R}$$

$$T = \frac{const}{\nu R V^{n-1}} \Rightarrow \ln \frac{T}{T_0} = \ln \frac{V_1^{n-1}}{V_2^{n-1}}$$

$$C_v \ln \frac{T}{T_0} = C_v (n-1) \ln \frac{V_1}{V_2};$$

$$\Delta S = C_v (n-1) \ln \frac{V_1}{V_2} + R \ln \frac{V_2}{V_1}$$

2) $\Delta U = \underline{-2666 \text{ Дж}}$

$\Delta S = \underline{-18{,}26 \frac{\text{Дж}}{\text{К}}}$

3) $Q - ?$ $Q = C \Delta T$, $C - C_p = n(C - C_v)$

$C(n-1) = nC_v - C_p$

$$C = \frac{3 \cdot \frac{3}{2} R - \frac{5}{2} R}{2} = \frac{9R - 5R}{4} = R$$

$T_1 = 240{,}67 \text{ К} \Rightarrow T_2 = 26{,}6 \text{ К} \Rightarrow$

$\nu R T V^{n-1} = const$

$\Rightarrow Q = \underline{-1779 \text{ Дж}}$

№ 4.44

Дано:	Решение:
$V_1 = 3$ л, $\nu_1 = 0{,}5$ $V_2 = 2$ л, $\nu_2 = 0{,}5$ $T_1 = 300$ К Найти: A_{max}	1) $Q = 0 = TdS = \Delta U + A$ $A_{max} = -\Delta U_{max}$ $U_1 = \frac{5}{2} \nu_1 R T_1$, $U_2 = \frac{5}{2} \nu_2 R T_1$

$U_н = 5 \nu R T_1$, $U_к = 5 \nu R T$

2) $S = const$, $S = S_0 + \nu C_v \ln \frac{T}{T_0} + \nu R \ln \frac{V}{V_0}$

$S = S_0 \Rightarrow 0 = \cancel{\nu_1} \left(C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_1 + V_2}{V_1} \right) + \cancel{\nu_2} \left(C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_1 + V_2}{V_2} \right)$

$\Rightarrow T_2 = 225\ K$

$3)\ -\Delta U = 5\nu R\,\Delta T = -5\cdot 0{,}5\cdot 8{,}31\cdot 75 = A = 1558$ Дж.

№ 5.32

Дано:

$F = -\frac{RT}{2}\ln(AT^3V^2)$

Найти: C_p

Решение:

$$1)\ C_V = \left(\frac{\partial Q}{\partial T}\right)_V = \left(\frac{T\,\partial S}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V$$

$$2)\ F = U - TS,\quad dF = -SdT - pdV$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V$$

$$\left(\frac{\partial F}{\partial T}\right)_V = -\frac{R}{2}\left(\ln(AT^3V^2) + \cancel{T}\cdot\frac{1}{\cancel{A}\cancel{T^3}\cancel{V^2}}\cdot\cancel{A}\cancel{V^2}\cdot 3\cancel{T^2}\right)$$

$$= -\frac{R}{2}\left(\ln(AT^3V^2) + 3\right) \Rightarrow S = \frac{R}{2}\left(\ln(AT^3V^2) + 3\right)$$

$$3)\ \left(\frac{\partial S}{\partial T}\right)_V = \frac{R}{2}\cdot\frac{1}{\cancel{A}\cancel{T^3}\cancel{V^2}}\cdot 3\cancel{T^2}\cancel{V^2}\cancel{A} = \frac{3R}{2T}$$

$$\Rightarrow C_V = \frac{3R}{2} \Rightarrow C_p = \frac{5}{2}R$$
